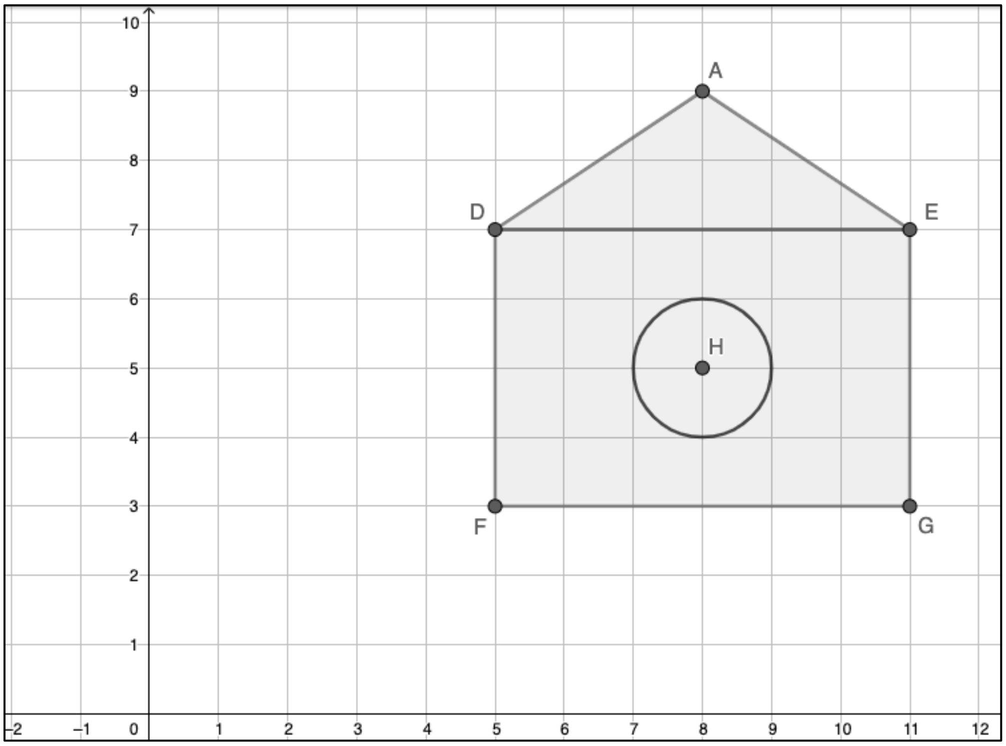
Geometry | Name Answer Key_________
Unit 13 | Date____________________
Lesson 14 Practice | Period__________________

Rory is designing the front of a birdhouse. The computer software generated coordinate points where each units equals an inch. The roof has vertices at vertices at $A(8,9)$, $D(5,7)$, and $E(11,7)$. The front entrance has a circle with a radius of 1 unit. The front wall has vertices at $D(5,7)$, $E(11,7)$, $F(5,3)$, and $G(11,3)$.



1. Classify the type of triangle that is on the roof?

   $DE = 6$

   $DA = \sqrt{3^2 + 2^2} = \sqrt{13}$

   $EA = \sqrt{3^2 + 2^2} = \sqrt{13}$

   $\triangle DAE$ is an isosceles triangle

2. What is the total area of the birdhouse?

   Area of Triangle $= \frac{1}{2}(6)(2) = 6$ in^2

   Area of Square $= (6)(4) = 24$ in^2

   Area of Circle $= \pi(1)^2 = 3.14$ in^2

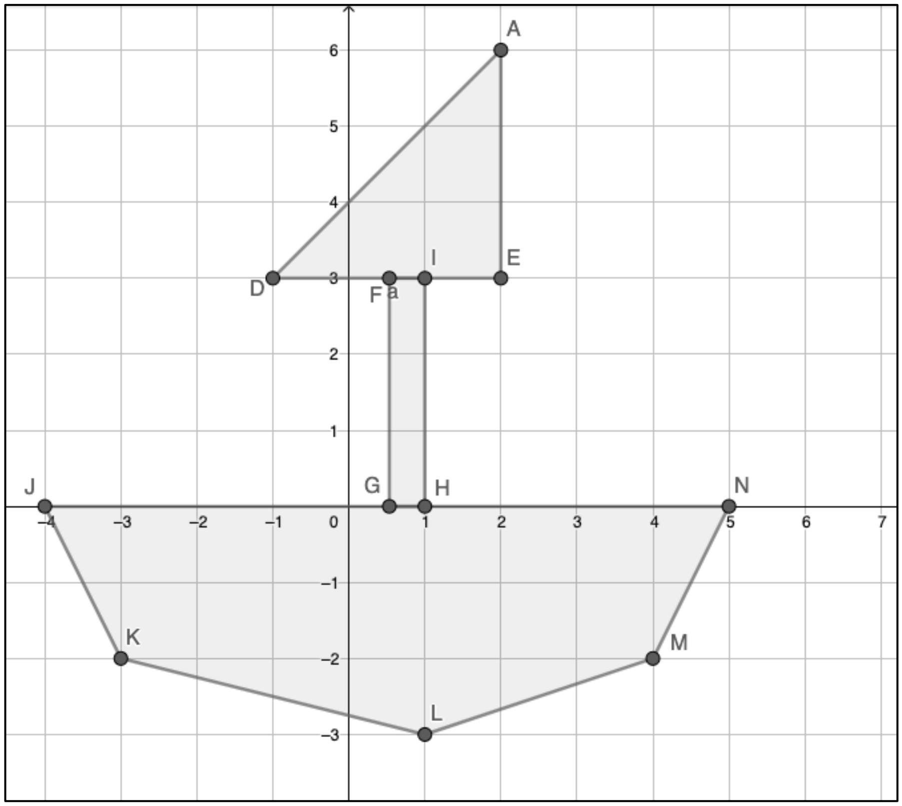
   Total Area $= 24 + 6 - 3.14 = 26.86$ in^2

3. Cartwright wants to use weather-stripping around the entrance. It costs $\$0.15$ per inch. How much weather-stripping is going to be needed around the entrance to the bird house?

   Circumference of Circle $= 2\pi(1) = 6.28$ in^2

   Cost of weather stripping $= 6.28 \cdot 0.15 = \$0.94$

Kima and Whuan are building a sailboat by hand. The layout of the sailboat is sketched on the blueprint shown.



4. If one square on the grid represents two feet, what is the distance around the sailboard on the blueprint?

$JN = 9$

$JK = \sqrt{2^2 + 1^2} = \sqrt{5}$

$KL = \sqrt{1^2 + 4^2} = \sqrt{17}$

$LM = \sqrt{3^2 + 1^2} = \sqrt{10}$

$MN = \sqrt{2^2 + 1^2} = \sqrt{5}$

$P = 9 + \sqrt{5} + \sqrt{7} + \sqrt{10} + \sqrt{5}$

$P \approx 20.76 \cdot 2 \approx 41.52$ ft

5. If one square on the grid represents three square feet, what is the total area of the sailboat on the blueprint?

Area of $DAE = \frac{1}{2}(3)(3) = 4.5$ units^2

Area of $JKMN = \frac{1}{2}(2)(9 + 7) = 16$ units^2

Area of $FGHI = (0.5)(3) = 1.5$ units^2

Area of $KLM = \frac{1}{2}(7)(1) = 3.5$ units^2

Total Area $= 4.5 + 1.5 + 16 + 3.5 = 25.5$ units^2 $= 25.5 \cdot 2 = 76.5$ ft^2